

RSES-Onco Introduction and Materials and Methods v0.11.1

RSES-Onco: a coverage-aware framework for prioritizing context-dependent cancer vulnerabilities from hidden functional enzyme backups

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Introduction

Synthetic lethality and related context-dependent genetic interactions offer a route to identify vulnerabilities that are selectively exposed by defined molecular states. In cancer, these relationships can reveal dependencies created by gene loss, altered pathway activity, metabolic adaptation or network rewiring [1,2]. Genome-scale CRISPR screening and resources such as DepMap have greatly expanded the systematic search for such dependencies, while also highlighting the need to distinguish general essentiality from context-specific effects [3-5].

Most computational and experimental strategies emphasize homologous paralogs, direct pathway partners or canonical genetic interactions. Functional redundancy, however, is not restricted to homologous proteins.

Non-homologous isofunctional enzymes (NISEs) can catalyze equivalent or closely related biochemical reactions despite limited sequence or structural homology. Such proteins may represent hidden alternatives that are missed by homology-centered searches and may become relevant only under particular cellular or tumor conditions.

The existence of the same nominal biochemical activity does not imply universal interchangeability. Enzymes can differ in expression, regulation, localization, cofactor usage, interaction partners, structural organization and response to genetic perturbation. Consequently, the biological relevance of a possible backup relationship may be restricted to a

functional microniche: the local and context-dependent combination of molecular and phenotypic evidence within which a directional relationship becomes plausible. In the present framework, a functional microniche integrates expression and compensation, coexpression and WGCNA organization, regulatory and protein-network context, dependency profiles, localization and biochemical/structural evidence.

Integrating these layers is complicated by unequal coverage. Some candidate pairs are richly annotated, whereas others lack structures, regulatory evidence, event-positive cell lines or sufficient samples. Treating missing evidence as zero can confound absence of data with negative biological evidence. Conversely, averaging only observed components without a coverage penalty can favor sparsely characterized candidates. A rigorous prioritization system therefore requires explicit eligibility rules, missing-data states, source-overlap controls and a coverage-aware mathematical formulation.

We developed **RSES-Onco**, a reproducible framework for prioritizing cancer-context hypotheses derived from curated human NISEs, homologous paralogs, collateral-loss relationships and all-target conditional-dependency discovery. RSES-Onco integrates DepMap dependency, copy-number and expression data; TCGA/GDC tumor-event matrices; STRING functional networks; DoRothEA/OmniPath regulatory relationships; Ensembl promoters and JASPAR/FIMO motif predictions; Human Protein Atlas localization; biochemical and structural annotations; and pharmacology or tractability evidence [6-18].

The framework separates evidence used directly in scoring from prioritization, external validation and interpretation. It also records source overlap so that multiple representations of the same original observation do not receive independent full weights. The resulting score is not presented as proof of clinical efficacy or synthetic lethality. Instead, it provides an auditable ranking of hypotheses for CRISPR validation, rescue experiments, expression perturbation, biochemical follow-up, structural interpretation and exploratory pharmacology.

Materials and Methods

Computational framework and reproducibility

All analyses were implemented in the SynLeth-RSES-Onco repository. The canonical execution protocol is provided in docs/END_TO_END_ARTICLE_PROTOCOL.md, and source-specific acquisition and provenance procedures are provided in docs/DATA_ACQUISITION_AND_REPRODUCTION_V0110.md. The software environment is defined by environment.yml and pyproject.toml. Publication assets are generated through scripts/run_publication_pipeline.sh.

The repository and publication framework are versioned as v0.11.0. The scientific ranking produced by the corrected scoring workflow is labelled RSES-Onco-expanded-v0.10.9, with expression/regulatory semantics eligibility-aware-wgcna-regulatory-v3.

Candidate-universe construction

Curated human NISE activities and proteins were used to construct directed NISE hypotheses. For each cross-group pair, both directional hypotheses were represented separately because loss of gene A followed by dependence on gene B is not assumed to be equivalent to the reverse direction. Ensembl Compara paralogs were incorporated as a distinct mechanistic class [11]. Additional candidates included curated benchmark relationships, collateral-loss hypotheses and all-target conditional-dependency discovery candidates.

Each candidate retained a stable pair identifier, lost/source feature, target gene, source class, relationship type, tumor-context eligibility and available source-specific identifiers. Composite biomarkers were preserved as composite events unless the source supported traceable expansion into atomic targets.

Cancer contexts

The primary disease contexts were colorectal, gastric and lung cancer. TCGA projects included COAD and READ for the colorectal context, STAD for the gastric context, and LUAD and LUSC for the lung context. Compatible DepMap cell lines were selected using the model annotation table. Actual release-specific model and tumor counts were recorded from the input files and were not forced to match previous releases.

DepMap inputs and validation

DepMap evidence included CRISPR gene-effect scores, whole-genome copy-number data, model annotations and protein-coding expression. The expected matrices were CRISPRGeneEffect.csv, OmicsCNGeneWGS.csv, Model.csv and OmicsExpressionTPMLogp1HumanProteinCodingGenes.csv. Input validation checked file availability, schema, model identifiers, gene-column mapping and cohort compatibility. Checksums were generated for release traceability.

TCGA/GDC tumor-event processing

Gene-level copy-number files were acquired from GDC using a reviewed ASCAT3 manifest for TCGA-COAD, TCGA-READ, TCGA-STAD, TCGA-LUAD and TCGA-LUSC. Downloads were validated by expected file size and checksum, and incomplete .part files were excluded. Gene-level data were aggregated into cancer-context event matrices, including context-specific homozygous-deletion tables.

Tumor-event supporting tables retained event definition, cancer, number of samples, frequency and source. Frequencies were restricted to biologically valid values. Any jitter used in plotting was applied only to graphical positions and was not stored as the true event frequency.

Dependency and selectivity analysis

DepMap gene-effect values were aligned to model annotations by ModelID. Candidate target dependency was evaluated within compatible cancer models and, where applicable, after stratification by the source/lost-gene event. General dependency and event-associated selectivity were treated as distinct concepts.

Supporting outputs retained the event-positive and event-intact sample sizes, group summaries, effect direction, effect-size estimate, nominal P value and FDR when computable. Candidates without sufficient compatible models were labelled as insufficient sample rather than assigned a zero component.

Expression and compensation

Expression values were obtained from the DepMap protein-coding expression matrix using the release-defined transformation. For each directional candidate and cancer context, the workflow retained expression of both genes, paired-model count, event-positive and event-intact summaries and pairwise expression metrics when available.

Spearman correlation and median absolute expression difference were used to characterize expression-profile divergence. Correlation alone was not considered proof of compensation. Compensation required an operational event-stratified pattern consistent with preservation or increased expression of the target after the relevant source-gene event. Non-computable comparisons remained missing.

Functional-microniche definition

The functional microniche represented the context-specific evidence environment of each candidate pair. It comprised six parent domains:

Functional-microniche domain	Weight
Expression context	0.20
Localization	0.15
Biochemical/structural evidence	0.15
Genetic phenotype	0.20
Interaction network	0.15
Regulatory network	0.15

Missing eligible components reduced coverage. Non-eligible domains were excluded from the eligible denominator.

Pairwise genetic-phenotype profile

Cancer-specific DepMap gene-effect profiles were compared using Spearman correlation, median absolute profile difference and divergence of dependency-positive model sets. Dependency-positive sets were defined by the threshold implemented in the source code, and set overlap was summarized by Jaccard similarity. The available divergence subcomponents were combined without replacing missing values by zero.

WGCNA expression-network analysis

Cancer-specific signed WGCNA networks were calculated separately for colorectal, gastric and lung models [7,8]. Candidate genes were supplemented with cancer-specific highly variable genes. Genes had to satisfy observation and non-zero-variance criteria. Remaining missing values were median-imputed only for network estimation, and the imputation count was recorded.

The primary correlation was biweight midcorrelation (bicor), with `maxP0Outliers=0.10` and `pearsonFallback="individual"`. Pearson was therefore used only for an individual gene or module eigengene with zero or non-finite median absolute deviation. The fallback was not applied globally and every affected entity was exported in an audit table.

The workflow selected a soft-thresholding power from the scale-free-topology diagnostics, calculated signed adjacency and signed topological overlap, performed hierarchical clustering and dynamic tree cutting, merged similar modules, and calculated module eigengenes, module membership (kME) and intramodular connectivity.

The cancer-specific WGCNA pair subscore combined:

WGCNA subcomponent	Weight
TOM divergence	0.40
Module divergence	0.30
kME divergence	0.30

Regulatory-network analysis

Regulatory evidence combined DoRothEA/OmniPath TF-target associations, cancer-specific TF-expression profile divergence and promoter-motif divergence [9,10]. The sublayer weights were:

Regulatory sublayer	Weight
TF-association divergence	0.40

Regulatory sublayer	Weight
TF-expression-profile divergence	0.35
Promoter-motif divergence	0.25

Canonical promoter windows were obtained from Ensembl. JASPAR vertebrate motifs were scanned with FIMO [11-13]. Motif predictions were labelled as sequence-based cis-regulatory support and were not described as direct binding, occupancy or causal regulation.

STRING functional-network evidence

STRING functional-network data were acquired with identifier mapping and resumable per-gene caching [10]. Evidence channels were retained where available, including experiments, curated databases, coexpression, neighborhood, cooccurrence, fusion and text mining. The STRING combined score was not interpreted as direct experimental interaction.

Localization evidence

Subcellular localization was derived from Human Protein Atlas source tables [14]. The localization component considered shared and distinct compartments, source reliability and compatibility with the proposed directional relationship. The presence of any annotation did not automatically produce maximum support.

Biochemical and structural evidence

Biochemical evidence retained molecular function, reaction, substrate, product, cofactor, domain and residue annotations when available. Structural evidence incorporated experimental structures and AlphaFold models, together with residue-level provenance from sources such as UniProt, M-CSA and PDBe [15-18]. AlphaFold confidence was not treated as experimental validation, and absence of an experimental structure was not treated as evidence of structural absence.

Pharmacology and tractability

Pharmacology and tractability evidence was assembled from sources including Open Targets, ChEMBL, DGIdb, MyChem, Pharos/TCRD and CIViC, with optional PRISM, GDSC and CTRP analyses [19-21]. Compound identity, mechanism, assay, model, response, development status, experimental tractability and clinical evidence were maintained as separate fields. Tractability was not equated with clinical efficacy.

Coverage-aware score

For candidate (i), context (c) and eligible domain set (E_{ic}), let (w_d) denote the domain weight, (x_{icd}) the normalized observed value and (m_{icd}) an indicator of observed usable evidence.

The observed eligible-domain score was:

$$S_{ic}^{observed} = \frac{\sum_{d \in E_{ic}} w_d m_{icd} x_{icd}}{\sum_{d \in E_{ic}} w_d m_{icd}}$$

Coverage was:

$$C_{ic} = \frac{\sum_{d \in E_{ic}} w_d m_{icd}}{\sum_{d \in E_{ic}} w_d}$$

The final adjusted score was:

$$S_{ic}^{adjusted} = S_{ic}^{observed} C_{ic} = \frac{\sum_{d \in E_{ic}} w_d m_{icd} x_{icd}}{\sum_{d \in E_{ic}} w_d}$$

The final parent-domain weights were:

RSES-Onco domain	Weight
Tumor event	0.16
Dependency	0.22
Selectivity	0.14
Expression compensation	0.08
Functional relation	0.06
Functional microniche	0.16
Validation and tractability	0.18

A real observed zero was retained as zero. Missing components were omitted from the observed denominator and penalized through coverage. Non-eligible components were excluded from both the observed and eligible denominators.

Source overlap and evidence roles

Evidence records sharing the same original publication, dataset or stable evidence identifier were grouped into one evidence unit. Multiple aggregators could confirm or annotate an observation but could not receive independent full weights for the same underlying event.

Evidence was classified as direct score evidence, prioritization evidence, independent validation evidence, overlapping/confirmatory evidence or interpretative evidence. This distinction was used to avoid double counting and circular validation.

Multiple testing and evidence categories

When multiple hypotheses formed a statistical test family, false discovery rate was controlled using the Benjamini-Hochberg method [22]. Nominal significance, FDR support and external validation were reported separately. A high integrated score was not interpreted as statistical significance, and statistical significance was not interpreted as clinical efficacy.

Candidates were classified as candidate-universe entries, computational hypotheses, prioritized hypotheses, hypotheses with microniche support, hypotheses with conditional-dependency support, nominally significant results, FDR-supported results, externally validated results, tractable candidates or candidates with clinical evidence.

Robustness analyses

The supporting workflow included raw versus adjusted score comparison, leave-one-domain-out analysis, individual component ablation, controlled weight perturbation, rank-correlation analysis, top-candidate stability, mechanistic-class and cancer-specific stability, and positive/negative control comparisons when supported by the available data.

Publication assets and quality control

All figures were generated programmatically. Each registered figure received an exact source-data TSV, generator script, input paths, reproduction command, PNG/PDF/SVG exports, layout-audit JSON and manifest entry. Tables received source and generator provenance through the table manifest and the human-readable asset reproduction registry.

The publication contract comprised 8 main figures, 69 supplementary figures, 231 image exports, 4 main tables and 44 supplementary tables. The documents were generated as editable DOCX files, rendered to PDF and page images, and subjected to automated and manual inspection. Supplementary Figures S68 and S69 were required to appear on separate pages.

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Promoter methylation analysis

Promoter methylation analysis

To incorporate epigenetic context without creating an independent evidence domain, open primary-tumor DNA methylation beta-value files were queried from the NCI Genomic Data Commons for TCGA-COAD, TCGA-READ, TCGA-STAD, TCGA-LUAD and TCGA-LUSC. The analysis used GDC-harmonized SeSAMe methylation-array outputs and the official GDC GENCODE-v36 probe annotations for HM27, HM450 and EPIC arrays.

Promoter-associated probes were defined using the official annotation as probes located from 2 kb upstream to 500 bp downstream of a transcription start site or classified in promoter/TSS-related groups. Masked probes were removed when the GDC general-mask field was available. Promoter methylation for each gene and sample was represented by the median beta-value across retained promoter probes.

For each directional candidate pair and cancer context, promoter beta-values for the hypothesized lost gene and target/paralog were matched within samples. Analyses required at least 10 paired samples. The workflow calculated median beta-values, median paired delta beta, Spearman correlation, a one-sided paired Wilcoxon test and Benjamini-Hochberg-adjusted P values. The directional promoter-methylation context score was defined as the median promoter beta of the hypothesized lost gene multiplied by one minus the median promoter beta of the target gene.

Promoter methylation was integrated inside the existing expression-compensation domain. The global expression-compensation weight remained 0.08, while its internal subweights were 0.70 for event-stratified expression compensation and 0.30 for promoter methylation context. The internal score was coverage-adjusted, so missing methylation reduced subcoverage but was not interpreted as biological zero. GDC methylation was applied only to the integrated TCGA-DepMap ranking and not to the DepMap-only ranking. Methylation beta-values were treated as epigenetic context and not as direct proof of gene silencing.

Methylation references

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